

GENERAL SPECIFICATION G3.4

## Ground Instrumentation and Monitoring

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## G3.4 GROUND INSTRUMENTATION AND MONITORING

### G3.4.1 General

- (a) The Building and Construction Authority (BCA) and other regulatory bodies require the installation and monitoring of instruments for excavations, tunnelling, piling and other construction works. Compliance with this Specification may not be sufficient to fully satisfy these regulatory bodies. Both the instrumentation required under this section and any additional instrumentation required to satisfy the regulatory bodies shall be designed and provided by the Contractor.
- (b) The board will engage an independent experienced Instrumentation and Monitoring Specialist/Firm (listed as a Specialist Builder for Site Investigation under BCA), to carry out instrumentation installation and monitoring works. The Contractor shall liaise with the appointed Instrumentation and Monitoring Specialist/Firm to install and monitor all the instruments which are requested by Contractor's Professional Engineer based on his ERSS design.
- (c) The Independent Instrumentation and Monitoring Specialist/Firm shall be report and be responsible directly to the Board or the S.O., for ensuring that the materials and workmanship conform to the standards stipulated in the Specifications, Land Transport Authority's Material and Workmanship Specification for Civil & Structural Works, and SS EN Codes or other Codes of Practice or Standards as specified.
- (d) The instrumentation shall be designed and selected to:
  - (i) Verify the assumptions made in the design;
  - (ii) Provide confirmation of the predicted behaviour of the support system during excavation and tunnelling;
  - (iii) Enable the assessment of the effects of the work on buildings, utilities and other structures;
  - (iv) Provide a record of performance;
  - (v) Enable construction to be carried out safely and soundly at every stage; and
  - (vi) Where required, enable appropriate contingency measures to be implemented in time.
- (e) To ensure the stability of structures in the vicinity of the proposed Works, the Contractor is required to implement a programme to monitor the surface and subsurface ground movements, variation of ground water table and pore pressure, and the settlement and distortion of buildings and structures.
- (f) The Contractor shall include in his Contract Price for the supply, installation, protection, maintenance, recording, monitoring and interpretation of soil instrumentation and survey and shall replace any defective, damaged or missing items, resulting from whatever cause, with new ones all at his own cost.

- (g) The Contractor shall provide all manpower and crew, as well as all plants, equipment and accessories, necessary for this work, all in the manner as may be directed by the S.O.. Equipment and instruments shall be installed to the manufacturer's instructions and shall be calibrated and tested as appropriate. Monitoring markers and devices shall be securely fixed in position. Due regard shall be given to the construction of the structure to be monitored and the layout of its primary support. The S.O. may also require settlement markers and inclinometers to be installed on or adjacent to structures that may be sensitive to settlements.
- (h) Within two months of Contract award, the Contractor shall submit to the S.O. for approval a proposed Action Plan, duly endorsed by his Professional Engineer (Geo), for instrumentation installation, monitoring, data handling and reporting. This Action Plan shall include full details of his proposed monitoring and reporting procedures. An essential part of the Action Plan shall be the consideration and preparation of contingency measures to prevent the development of unacceptably high ground and structure movements.
- (i) The Contractor shall be solely responsible for determining the locations and types of geotechnical instruments to be installed on site. The actual instrumentation locations will be chosen to accommodate construction operations, subject to the review and approval of the S.O.. The Contractor shall note that the types of soil/geotechnical instruments as per BQ are the minimum requirement that must be provided under the Contract. The contractor shall propose the whole soil movement monitoring plan with the type, number and location of each monitoring instruments clearly indicated in their proposal and subject to relevant authorities and S.O.'s approval. Additional monitoring stations, devices and procedures shall be provided if deemed necessary or if so directed by the S.O. during the progress of the Works.
- (j) The Contractor shall submit detailed drawings with his P.E. (Geo.) endorsement showing the precise locations and depths for instrument installations for the approval of the S.O. within two months from the Date of Commencement and at least 21 working days prior to installation.
- (k) The Contractor shall submit detailed method statements for the installation of all instruments for S.O.'s approval within two months from the date of commencement and shall be no less than 21 working days prior to installation.
- (l) The method statements shall include all relevant details and drawings of the instruments, manufacturer's specifications and recommendations for installation and maintenance, details of drilling equipment and drilling methods, details of all materials (including samples), details of particular sequencing of assembly and connections, details of any backfill, proposed grout mixes, details of monitoring arrangements and verification procedures.
- (m) The method statement shall also include a programme of installation related to the main construction activities, subject to the approval of the S.O..

- (n) Instrumentation plans shall be prepared at a scale not greater than 1:500 clearly indicating the location and installation height/depth of all instrumentation. Sectional drawings shall show the extent and depth of permanent and temporary works and the strata where the instruments are to be installed. All instruments shall be clearly identified in an alphanumeric sequence. Each drawing shall show all of the instrumentation to be installed within the area covered in the drawing. Where multi-sensor instruments are used, they shall be clearly identified with the reduced level and/or direction of each individual sensor, such information is to be provided either on the drawings or in separate tables. Where buildings, structures or utilities subject to special protection works are encountered, a detailed drawing shall be prepared for each individual building, structure or utility indicating the exact location of all monitoring points, including the reduced level of the proposed instrument. Sections shall also be prepared identifying clearly the extent of any existing features, cracks etc., requiring monitoring. Where possible the structural layout of the building shall also be indicated on these drawings. All drawings for buildings, structures or utilities subject to special protection works shall be prepared at a scale not greater than 1:50 except where prior approval has been granted by the S.O..
- (o) The Contractor shall be responsible for determining the maximum predicted values of total and differential settlement for both the ground surface and those of buildings, structures and utilities etc., considered by the S.O. to be at risk of damage due to the Works including tunnelling, pipe jacking and shaft excavation. These predictions of settlement shall be used to establish the safety limits for soil/geotechnical movements in terms of total and differential settlement to ensure that all structures and properties are not endangered.
- (p) The settlements and suspension limits shall be determined by rigorous settlement analyses using recognized methods and submitted to the S.O. for approval. The safety limits shall be submitted to the S.O. for information. The S.O. reserves the rights to improve on the safety limits when deemed necessary. Surface settlements shall not exceed the safety limits.
- (q) In monitoring the instrumentation, the Contractor shall employ a two-level "Trigger Value" system. These Trigger Values shall be named as "Alert" and "Work Suspension" Levels as per BCA's latest Advisory Note.
- (r) The Alert Level shall be reached when the observed measurements at any monitoring station reach 70% of the safety limit and is accompanied by a trend that indicates the predicted value will be exceeded prior to the completion of the Works. Upon reaching the Alert Level, the Contractor shall submit a review of the following activities to the S.O. with a recommendation for the most appropriate action to arrest the trend of ongoing movement:
  - (i) Adequacy of instrumentation;
  - (ii) Frequency of monitoring;
  - (iii) Construction methodology and sequence; and
  - (iv) Requirement for preventive measures.

- (s) The Work Suspension Level shall be reached when the observed measurements at any monitoring station reach 100% of the safety limit and is accompanied by a trend that indicates the predicted value will be exceeded prior to the completion of the Works. Upon reaching the Work Suspension Level, the Contractor shall immediately suspend all work and notify the S.O. and propose a rectification plan endorsed by the P.E. (Geo.) to arrest the trend of ongoing movement. The rectification plan shall be submitted to the S.O. for approval no more than 24 hours after the monitor reaches the Action Level.
- (t) The Contractor shall calibrate and maintain in good working condition all monitoring instruments and logging devices throughout the Contract period and regularly check the integrity of all temporary bench marks or base reference stations. If in the opinion of the S.O., any instrument or device is not functioning properly or accurately, it shall be replaced immediately at the expense of the Contractor.
- (u) The Contractor shall carry out regular monitoring to frequencies agreed by the S.O. and provide the S.O. with suitable records on a regular basis, throughout the duration of the Works. The frequency of monitoring the instruments and reporting to the S.O. shall be as follows or at a lower frequency as determined by the S.O.. Payment of this item shall therefore not exceed the amount based on the frequency of monitoring stated below. The Contractor shall however, if he deems necessary, monitor at closer intervals at his own expense.
- (v) The Contractor shall observe record and analyse the readings to establish trends in movement and shall make available results to the S.O. in accordance with an agreed programme. During the execution of the Works, defects which arise during the course of the Works shall be recorded. The Contractor shall keep records of such inspections and a copy shall be available to the S.O..
- (w) All readings and measurements shall be properly taken, recorded and interpreted by the Instrumentation and Monitoring Specialist/Firm's PE and, endorsed by the Contractor's P.E. (Geo) and two sets of the report shall be submitted to the S.O. weekly. It shall be clearly understood that the submission of the reports on monitoring to the S.O. are for the S.O.'s information. The Contractor shall be solely responsible for the analysis and interpretation of all the readings and measurements and for taking all rectification measures where necessary. When abnormal readings or measurements are obtained, the Contractor shall immediately stop all movements or changes, and shall immediately investigate the causes of abnormalities. The Contractor shall take all necessary rectification measures before resuming each activity, all at the Contractor's own cost.
- (x) Monitoring of settlement and defects arising during the course of the Works shall continue at agreed intervals for a period of at least 6 months after completion.
- (y) The Contractor shall allow in his rates for carrying out the monitoring programme as specified, including the service of the Instrumentation and Monitoring Specialist/Firm and any additional stations, devices and procedures as may be directed by the S.O. or the client.

- (z) The Contractor shall note that as-built drawings may not contain building / foundation / utilities information for every single building / foundation / utility that is likely to be affected by the Works. Where such information is lacking, the Contractor shall be responsible for seeking and ascertaining the relevant information.

#### **G3.4.2 Buildings, Structures and Utilities to be Assessed**

- (a) The Contractor shall take all necessary steps to ensure that the stability of structures in the vicinity of the proposed Works are maintained and shall avoid causing any unacceptable damage to roads, buildings, structures, utilities, trees or any other features during the period of the Contract.
- (b) All buildings, structures and utilities likely to be affected by the Works shall be assessed based on local design criteria such as Land Transport Authority's Civil Design Criteria Specification for Road and Rail Transit Systems for the likelihood of damage. Based on Contractor's detailed construction methodology and predicted ground movements, he shall determine the zone of influence of the Works. The Contractor shall carry out visual inspections of buildings, structures or utilities as required for his purpose of carrying out the damage assessment study.
- (c) For buildings on mixed foundations (i.e. part of the structures is founded on piles and part on shallow foundations), the Contractor is required to:
  - (i) Prepare and submit report on the as-built structural and foundation details of the building;
  - (ii) Carry out inspection of the building to verify if there is any structural connection between parts that are founded on different foundation system; and
  - (iii) Separate the structural connection (if any) between parts on different foundation system and implement any associated strengthening works.

#### **G3.4.3 Settlement Analysis Report**

- (a) The Contractor shall submit a Settlement Analysis Report within two (2) months of Contract award to document the assessment of expected ground movements and the response of the adjacent buildings and structures to the execution of his work.
- (b) The Settlement Analysis Report shall include, among others, the following:
  - (i) Geotechnical characteristics of the ground;
  - (ii) Proposed methods and sequencing of construction and types of equipment to be used;
  - (iii) Type of structures including the existing conditions of the structures and foundations together with allowable total settlement and foundations together with allowable total settlement, differential settlement, angular distortion and/or strains. These values shall not result in a damage category that exceeds the requirements of the Land Transport Authority's Civil Design Criteria Specification for Road and Rail Transit Systems;
  - (iv) Set trigger levels for each stage of the Works together with contingency plans and actions when review levels are breached;

- (v) Assumptions and calculations on which the assessments of the ground movements, movement of structures, etc., are made;
  - (vi) Expected ground movements (both horizontal and vertical) during the execution of the Works, including drawings showing predicted settlement including settlement contours. The assumptions made in respect of the temporary works schemes are to be shown in the drawings;
  - (vii) The Contractor's proposal for additional site investigation where required for finalization of settlement and protective works design;
  - (viii) The risks of predicted settlements being exceeded and mitigation measures to reduce these risks;
  - (ix) Contingency proposals for protective works, if predicted settlements are exceeded; and
  - (x) The Contractor shall also carry out an assessment of the effect of the predicted short and long term ground and structural movement on all adjacent buildings and structures and identify, if any, requires protection works. The Contractor shall develop the design of the protection works.
- (c) The Contractor's Settlement Analysis Report shall follow the format of a Preliminary, Pre-final and Final Settlement Analysis Report, taking on any additional soil investigations or other information obtained. Each report (preliminary, pre-final and final) should highlight any changes with full justification for the modifications. Protective works design for all structures must be fully developed. The Final Settlement Analysis Report must be submitted prior to the commencement of Works. No work will be allowed until the Final Settlement Analysis Report is accepted by the S.O..

#### **G3.4.4 Protective and Corrective Measures**

- (a) Based on his assessment, the Contractor shall propose and install any protective works, which shall be established by him subject to the acceptance of the S.O.. Protective works shall be in the form of modification to support systems and foundations, ground improvements, compensation grouting, jacking or structural strengthening, groundwater control systems, propping, shoring or other methods proposed by the Contractor. Care shall be taken to ensure that the selected methods of preventive measures does not have adverse effects on the buildings/structures/utilities and has a proven record for this type of construction in similar ground conditions.
- (b) The Contractor shall also propose contingency plans subject to acceptance of the S.O. and implement these in the event that the measured movements exceed the accepted trigger levels or indicate that final long term movement will exceed the allowable values established in the Final Settlement Analysis Report.
- (c) For every building identified as requiring protective works, the Contractor shall submit design calculations, working drawings, method of working and programme at least four (4) weeks before the commencement of any work (including the protective measures) in the area of the building. No work shall be carried out prior to the acceptance of the proposal by the S.O..

### **G3.4.5 Groundwater Control Systems**

- (a) Before the commencement of the Works, the Contractor shall provide full details of his proposals for groundwater control systems for the S.O.'s acceptance. The full details shall include, but not be limited to the following:
  - (i) Predicted groundwater drawdown detail relating to Contractor's sequence and method of construction;
  - (ii) Proposed groundwater control system and its efficiency;
  - (iii) Locations of any elements of systems for relief of pore water pressure or recharge wells and location of standpipes inside and outside the Works;
  - (iv) Impact assessment and protective measures for any building, structure, infrastructure or utility due to groundwater control;
  - (v) Redundancy in the system to cater for breakdown or failure of any element of the groundwater control system; and
  - (vi) Proposed contingency measures for groundwater control.
- (b) Control of groundwater may be used in order to ensure stability of excavations and to limit groundwater drawdown resulting from seepage into excavations. However, if in the opinion of S.O. these activities would have an adverse impact on any road, slope, structure or utility, such operations will not be implemented.
- (c) Groundwater control systems beneath the excavation bottom may consist of either active or passive relief wells subject to the acceptance of the S.O.. Such systems shall be designed and implemented to ensure the stability of the temporary earth retaining structures at all stages of construction.
- (d) Notwithstanding the requirements to provide an essentially watertight retaining wall structure in the vicinity of compressible soils, groundwater control measures are to be used to control seepage where the Contractor predicts that groundwater drawdown or seepage related consolidation settlements occur.
- (e) Where recharge systems are proposed, the design of these systems shall take account of the following requirements:
  - (i) The recharge system is to be installed and in operation prior to the commencement of Works;
  - (ii) Each recharge well boring shall be logged and the recharge well designed to suit identified conditions;
  - (iii) The well system shall be arranged to ensure that recharge occurs relatively uniformly where wells are assumed to be normally operational for 90% of the time, not including contingency well pump/well failure;
  - (iv) The design injection pressure shall ensure that the wells and surrounding ground remain stable and do not heave at the surface;
  - (v) Well screens that are installed must be durable over the duration of the project; and
  - (vi) Recharge wells shall be periodically back-flushed using a submersible pump to clean out the well screen section to ensure the well efficiency.



#### **G3.4.6 Monitoring and Recording of Underground Work**

- (a) The Contractor shall survey, monitor and record pipe jacking, tunnel and shaft excavation construction as it proceeds to form a record of the work. Monitoring shall generally be per unit of advance and include line, level, cross-sectional accuracy, shift advance and total advance.
- (b) Where shields and machines are employed, the Contractor shall monitor per unit of advance, the altitude of the shield or machine. Information to be recorded includes rate of advance, line, level, square, plumb, and roll. Where applicable, face pressure, slurry density, slurry viscosity, slurry level, slurry flow, cutter speed, rotation direction, torque screw conveyor speed, air pressure, and to the degree of accuracy applicable, volume of material excavated.
- (c) When pipe jacking, the Contractor shall also monitor and record the jacking forces and the quantity, type, location and pressure of injected lubricants.
- (d) All information recorded by the Contractor shall be provided to the S.O. on a daily basis unless another interval has been agreed.
- (e) Where the Contractor considers that any corrective action he may take will exceed the tolerances in the Contract, he shall so inform the S.O. and obtain his agreement.
- (f) All significant groundwater ingress shall be recorded and monitored.
- (g) All atmospheric testing shall be recorded and monitoring for all gases carried out in accordance with BS 6164.
- (h) The Contractor shall keep copies of all recent face records at the workplace for the information of supervisory personnel.
- (i) When constructing segmental lining, the Contractor shall undertake survey checks pertinent to the accurate erection and position of segments during each ring built. The relative attitudes of the lining and the tunnelling machine/shield shall be recorded.

#### **G3.4.7 Instrumentation and Monitoring Requirements**

- (a) The Contractor shall provide the specialist contractor access for him to carry out installation of instruments and allow for the monitoring of the works in a timely and safe manner.
- (b) Sufficient time shall be allowed between installation of instruments and commencement of relevant site activities to enable a reliable set of base readings to be established. These timescales are to be agreed with S.O..
- (c) The Contractor shall coordinate with the specialist contractor to ensure that any of the installed instruments is not damaged on site. Any damaged instrument shall be replaced by the specialist instrumentation contractor at the cost of the Contractor.

- (d) All monitoring instruments shall be manufactured by companies with proven experience in the field of construction or geotechnical instrumentation. Materials, design and construction shall be of the highest quality to provide robust, corrosion and vibration resistant instruments. Cabling and connectors for instrumentation shall be of robust and water resistant construction, and shall be corrosion resistant. All instrumentation shall have a satisfactory proven working life and be new at the time of installation/contract commencement. The accuracy and dependability of the instrumentation shall not be significantly affected by changes in temperature, humidity, stray currents or contaminants that may be encountered.
- (e) The Contractor shall be solely responsible for conducting stringent monitoring programme (with the advice of Professional Engineer and qualified persons engaged by the Contractor) for the soil/geotechnical instrumentation and shall submit suitable records and reports to the S.O.. The Contractor shall be solely responsible for ensuring that the frequencies of monitoring the instruments are adequate to detect immediately all soil/geotechnical movements or changes for the safe execution of the Works without endangering any structures or properties. All readings and measurements shall be properly taken, recorded and interpreted by qualified persons and one (1) set shall be submitted to the S.O. immediately.
- (f) The Contractor shall provide the following indicative instrumentation and monitoring requirements for excavation, pipe jacking and tunnelling works:
  - (i) Inclinometers installed in the ground to monitor lateral ground deformation. Provide plastic type with outside diameter of 70 mm. Inclination not greater than 1:150;
  - (ii) Tiltmeters;
  - (iii) Precise survey prisms;
  - (iv) Vibrating wire piezometers to measure pore pressure changes. Provide heavy-duty steel box with lock;
  - (v) Settlement survey of surrounding ground using precise levelling surface settlement points and settlement plates;
  - (vi) Standpipes to monitor groundwater levels. Provide UPVC tubing wrapped in a geotextile;
  - (vii) Extensometers; and
  - (viii) Crackmeters.
- (g) All equipment shall be installed and tested in accordance with the manufacturer's instructions or recommendations. Testing shall be undertaken, as necessary, to ensure satisfactory functioning of the equipment at each stage of the installation. Instruments found to be malfunctioning at any time shall be reported to the S.O. immediately and replaced at the earliest opportunity, but in any case no later than 24 hrs from the time of the fault being identified.

- (h) The Contractor shall note that all the instruments are to be delivered with instruction manuals and calibration data. The instrumentation details and proposed plans shall be submitted to the S.O. before installation. They shall be installed and functional prior to the commencement of the excavation work. Under no circumstances will excavation be allowed to carry out unless the necessary soil/geotechnical instruments have been installed and become functional such that the stability of the embankments can be monitored during the excavation. The contractor shall provide adequate protection, staging for monitoring and access to these working conditions at all times and immediate reinstatement shall be carried out upon the damage of any of these instruments to ensure continuation of monitoring.

#### **G3.4.8 Minimum Monitoring of Utilities**

- (a) All gas, water, sewers, electrical cables, drains, etc., within the monitoring zone proposed by the Contractor and approved by the S.O. shall be monitored with settlement points placed onto or just above the utility, at intervals of not more than 25m.
- (b) Utilities that are less than 4 m below ground level should be monitored with points on the utility, while utilities more than 4m below ground level can be monitored with points placed 2 m above the utility. Joints in electrical cables with voltage higher than 22 kV and in fibre optical systems shall be monitored for movement across the joint.

#### **G3.4.9 Monitoring of Areas of Ground Treatment**

- (a) Where local ground treatment/soil stabilization works are performed, a local grid of surface settlement monitoring points shall be established over the treatment area. The spacing between points shall not be more than 10 m in either direction and shall extend at least 10 m from the edge of the treatment zone in all directions.
- (b) Similar local arrays shall be established at all utility or structure within 10 m of the treatment zone.

#### **G3.4.10 Instrumentation Equipment**

- (a) No work shall commence until the instrumentation programme has been approved by the S.O. and all necessary instruments are properly installed, calibrated and functioning properly.
- (b) The minimum monitoring for excavations of depths 4 m or deeper and tunnels including pipe jacking shall be based on the latest Land Transport Authority's Civil Design Criteria.
- (c) The Contractor shall be required to carry out additional monitoring work including but not limited to additional instrument points in critical situations where the S.O. deems that the safety of the Works or surrounding properties or utilities or the public is at risk. Under such circumstances, the Contractor shall carry out monitoring work at a frequency of twice a day and to mobilize additional suitable equipment and trained personnel within 24 hours of receiving instruction from the S.O.. No claims for extra cost or time incurred by the Contractor as a result of such additional monitoring work shall be allowed.

- (d) If at the later stages of construction, the S.O. is in the opinion that the monitoring is no longer essential, the minimum frequency of monitoring may be decreased, and readings may be taken at longer time intervals subject to written approval of the S.O..

#### **G3.4.11 Frequency of Readings**

- (a) The frequency of readings for the various instruments throughout the duration of the Contract shall be:

|                                |                                                |
|--------------------------------|------------------------------------------------|
| Piezometers                    | Readings shall be taken at least once daily    |
| Inclinometers                  | Readings shall be taken at least once daily    |
| Standpipes                     | Readings shall be taken at least once daily    |
| Settlement Markers/Plates      | Readings shall be taken at least once daily    |
| Tiltmeters                     | Readings shall be taken at least thrice weekly |
| Extensometers                  | Readings shall be taken at least thrice weekly |
| Crackmeters                    | Readings shall be taken at least thrice weekly |
| Prisms                         | Readings shall be taken at least thrice weekly |
| Vibration monitoring equipment | Readings shall be taken at least once daily    |

- (b) The frequency of the readings for the various instruments shall be increased to at least once daily when the instruments are within the influence zone of the tunnelling works and when alert levels of the instruments are reached and to comply with the latest requirements of regulating Agencies. The Contractor is deemed to have allowed in his price for complying with the requirements for this Clause. No claims whatsoever in this respect shall be entertained by the Board.
- (c) If instrument readings have remained stable for a period of four consecutive weeks, at the discretion of the S.O. readings may continue on a fortnightly basis. If variations in the fortnightly readings are observed, readings shall return to a weekly basis or such other frequency agreed by the S.O..
- (d) Under no circumstances shall the data obtained from the above be used by the Contractor for the purposes of extra payment or extension of time. All data obtained shall become the property of the Board.

#### **G3.4.12 Recording Equipment and Ancillaries**

- (a) Instrumentation Cabling
- (i) All cables used in enclosed areas shall be low-smoke zero halon cables;
  - (ii) Cabling and tubing shall be provided with sufficient slack to accommodate potential ground movement;

- (iii) Cabling for the instrumentation shall be neatly and securely fixed to appropriate cable trays, which in turn shall be securely fixed to the structure;
  - (iv) Cabling and connectors for instrumentation shall be of robust and water resistant construction and shall be corrosion resistant. Any non-metallic items shall be extinguishable in the event of fire and emit no toxic fumes; and
  - (v) Each cable shall be tagged at least at the instrument end connection and at the logger connection. A schedule shall be maintained showing all cables and their respective connections.
- (b) Terminal Boxes for Remote Readout
- (i) Terminal boxes shall collect the cabling from the various remotely read instruments and from local junction boxes, so that monitoring can be conveniently carried out from a single position close to an instrumented section.
  - (ii) Terminal boxes shall be positioned for ease of access and shall be protected from damage.
  - (iii) The terminal and junction boxes shall be of robust water-resistant metal construction. The boxes shall have lockable sealed doors. All cable entries to the boxes shall be through watertight seals.
- (c) Remote Readout Facilities
- (i) All instruments with electrical sensors shall be capable of being read remotely using portable readout units connected via the terminal boxes. Readout units shall be supplied together with leads for plugging into the measuring device and terminal boxes.
  - (ii) Readout units shall have facilities for recording monitoring data onto a suitable medium for subsequent downloading to a computer for processing.
  - (iii) For critical instruments that require real time monitoring, modem link and connection for direct computer link shall be made to the S.O.'s site office.
- (d) Data Loggers
- (i) Data loggers shall collect the cabling from the various remotely read instruments and from junction or terminal boxes.
  - (ii) Data loggers shall be reliable and compatible with the instruments and have back-up power facilities. The reliability and compatibility of the logger shall be proven prior to installation, by demonstration and records of previous works.
  - (iii) All data loggers shall be connected to electrical mains by a flex outlet without switches. Plug sockets shall not be used.
  - (iv) Data loggers shall be:
    - 1) Positioned for ease of access;
    - 2) Programmable to log data at any specified interval, including real-time;
    - 3) Capable of storing data for 14 days;
    - 4) Easily connected to a portable computer directly by cables or by modem link;

- 5) Unaffected by external electromagnetic influences, and shall not affect other electrical equipment;
- 6) Insulated from rain and dust;
- 7) Fused against electrical power surges; and
- 8) Capable of automatic shut down in the event of overheating or short-circuiting.

#### **G3.4.13 Settlement Points**

- (a) Adequate surface settlement points shall be provided to monitor the influence of shaft and tunnel excavations, and pipe jacking on surrounding ground and buildings, factories, key utilities and road structures.
- (b) A grid of surface settlement points shall be established along the alignment of the pipeline or tunnel. The grid shall consist of points at 20 m intervals perpendicular to the centre-line of the proposed pipeline or tunnel. Each line shall consist of a minimum of 5 points at 5 m centres with the middle point on the centreline of the tunnel.
- (c) These settlement points shall be surveyed by the use of precise survey instruments and at least 3 permanent bench marks shall be established on stable and firm ground away from the working area located outside the zone of influence of the Works and not subject to ground movement for this purpose.
- (d) Settlement points shall also be established to monitor the settlement of all gas pipes, water pipes, sewers, electrical cables and fibre optical systems. For shallow services less than 4 m depth settlement points should be established where possible directly on the utility or by agreement with the S.O. on the surface on the line of the utility. For deep services, greater than 4 m depth, settlement points shall be established within 2 m of the top of the utility. Settlement points on utilities shall be established at a maximum spacing of 25 m.
- (e) As the excavation proceeds, all surface settlement points within an area of 100 m ahead and 100 m behind the ends of the trench shall be surveyed daily. Outside this area, selected points at intervals as determined by the Contractor's P.E. shall be surveyed at least once a week.

#### **G3.4.14 Water Standpipes**

- (a) Standpipes shall be installed in boreholes as instructed by the S.O. or his representative for recording the highest ground water level encountered and changes in this level.
- (b) Standpipes shall consist of steel or UPVC tubing of at least 25 mm diameter. Slots or holes shall be formed in the lower 1.5 m of pipe, or porous element shall be provided. The lower end shall be plugged to prevent entry of soil during installation and the total length of each standpipe shall be recorded.
- (c) A layer of suitably graded filter material shall be placed on the bottom of the borehole prior to installation the standpipe.
- (d) The boring shall be backfilled with clean gravel or coarse sand to within 1m of ground level unless otherwise specified, and thereafter with impervious material so as to prevent the entry of surface water.

- (e) Approved arrangements shall be made by the Contractor to protect the top of the standpipe.
- (f) The ground water level shall be recorded immediately before and after the installation of the standpipe and the sounding device shall also be used to check the total length of standpipe installed.

#### **G3.4.15 Vibrating Wire Piezometers**

- (a) The type of tip shall be chosen to suit the expected maximum groundwater pressure and the characteristics of the surrounding ground.
- (b) The combined accuracy of instrument and readout device shall be within +1.0 % of the true pressure.
- (c) Installation in Boreholes Drilled from the Surface:
  - (i) All piezometers shall be tested before installation by submergence in still clean water to a series of depths appropriate to the anticipated pressure range of operation in order to verify the calibration and accuracy of the instrument.
  - (ii) The tip of each piezometer shall be air free and the porous stone of the tip shall be fully saturated by submergence in clean de-aired water for a period of at least 24 hours prior to installation. Complete de-airing of the porous stone shall be confirmed by periodic weighing.
  - (iii) Each piezometer shall be installed in a separate borehole of 100mm nominal diameter unless otherwise accepted by the S.O.. The complete installation shall be carried out as soon as practicable, and no later than 24 hours, after drilling to minimize the amount of deterioration or alteration of the ground at the location of the piezometer tip.
  - (iv) Drilling to the required depth shall be carried out without the use of airflush in the vicinity of the tip position. Casing shall be used to stabilize the hole through unstable ground.
  - (v) The hole shall be flushed until it is filled with clean water. The piezometer tip shall then be inserted under water and backfilled with clean sand to form a sand cell. The sand cell shall be of 1 m height with the piezometer located in its centre.
  - (vi) A plug of bentonite formed of either pellets or chips shall then be tamped over the sand to form a minimum thickness of 1m. The hole shall then be backfilled with bentonite/cement grout mix to the acceptance of the S.O.. Grout backfill shall terminate at the base of the instrumentation chamber.
  - (vii) Where accepted by the S.O. the Contractor may use 'push-in' vibrating wire piezometers utilizing a mandrel of not less than 0.5m in length. In such cases there will be no requirement for a sand cell. Backfilling using bentonite and grout as specified shall be used.
  - (viii) Casing, if used to maintain the hole open, shall be withdrawn carefully so as not to damage the instruments or cables, and concurrently with the progress of the backfilling.

- (ix) Accurate records of the depths of the piezometer, sand cell and bentonite seals shall be kept, and readings shall be recorded at each stage of the installation. For push-in piezometers the increase in pore pressure as well as dissipation shall be recorded using a data logger. The maximum pressure allowable during installation shall be 50% of the maximum for which the piezometer is rated.

#### **G3.4.16 Inclinometers**

##### **(a) General**

- (i) Inclinometers shall take the form of an access tube with four key ways (in pairs at right angles) which shall be grouted in a borehole. The orientation of the key ways shall be such that they are parallel and perpendicular to the orientation of any tunnel, excavation or wall or otherwise as accepted by the S.O..
- (ii) When taking readings, the inclinometer probe shall be inserted in the tube and shall travel along the whole length of the tube with its wheels engaged in each pair of the key ways in turn to obtain deformation measurements in both directions. The probe wheels shall be compatible with the key ways. Changes in inclination shall be displayed on a portable readout unit.
- (iii) A suitable means of measuring the position of the probe down the borehole to an accuracy of  $\pm 10$  mm from a reference point at the top of the borehole shall be provided.
- (iv) Both the probe and the portable readout units shall be calibrated at intervals recommended by the manufacturer.
- (v) The inclinometer tubing shall have an outer diameter of 85 mm and shall be pre-grouted into a minimum 125 mm diameter borehole or a steel duct of at least 150mm cast into any pile or wall.
- (vi) The inclinometer tubing shall be of uniform section and be free of twist in manufacture and shall not be twisted during installation. The tubing shall be provided with appropriate couplings for joining tube lengths to the required borehole length. The base of the tube shall be capped. Cap and coupling joints shall be adequately fixed and sealed so that the tube is grout and dirt tight. The grooves shall be consistent in cross section. The tubing material shall be ABS plastic unless otherwise accepted by the S.O..
- (vii) The joined and sealed tubing shall be concurrently filled with clean water while being lowered into the borehole and the annulus grouted using a bentonite/cement grout tremied to the bottom of the hole. The tubing shall be held in place at the bottom until the grout sets. The tubing shall not be held down with weights attached to the top.
- (viii) Immediately following installation a spiral and inclination check shall be carried out. Corrections required to data readings shall be determined and applied.
- (ix) During earthworks operations the inclinometer tubes shall be extended in lengths and ahead of the fill to a manner accepted by the S.O..



- (x) At the final ground level, the inclinometer tubing shall be fitted with a removable cap protected by a manhole cover and surface protection barrier.
- (b) Electrolevel Inclinometers (Remotely Read)
  - (i) The inclinometer shall consist of a string of biaxial places electrolevels located in a slotted plastic inclinometer.
  - (ii) Each electrolevel shall consist of waterproof sensors remotely monitored using a computer data logging system.
  - (iii) The resolution of the electrolevel shall be 2 seconds of arc or better and the measuring range shall be at least + 9 degrees. The operating temperature range shall be at least 0°C to +50°C.
  - (iv) Calibration certificates for each electrolevel showing calibration for both tilt and temperature shall be provided. Both calibrations shall fully characterize the electrolevel output response over both the specified tilt and the operating temperature range. Sample calculations to translate output to movement shall be provided.
  - (v) The electrolevels shall be installed at spacings not greater than 3 m and shall be connected with a rigid rod or beam to give a fixed gauge length. At the end of the sensor and the gauge extension a spring loaded wheel assembly shall be fitted. Sensors mounted on sleds will not be acceptable. Stability and gain of the reading system shall be checked at weekly intervals by means of three "dummy" electrolevels using precise resistors.
- (c) Inclinometers (Manually Read)
  - (i) A mechanical pulley shall be used for taking all readings.
  - (ii) The datum readings should be established using at least 2 probes and shall be submitted together with the serial numbers of the probes.
  - (iii) All subsequent readings shall only be taken using probes with which the datum readings have already been established.
  - (iv) Calibration certificates for each inclinometer probe shall be provided to the S.O..
  - (v) The system accuracy shall be  $\pm 6$  mm per 25 m of casing or better.
  - (vi) The stability of the reading system shall be checked at weekly intervals through the use of checksum calculations.
  - (vii) The inclinometer and logging system shall be capable of recording movements in both directions (i.e. biaxial).
  - (viii) The probe used shall have the following:
    - 1) A wheel base of 500mm or greater;
    - 2) A measurement range of  $\pm 35$  degrees or greater from vertical;
    - 3) A resolution of 0.02mm or better per 500mm with a repeatability of  $\pm 0.1\%$ ;
    - 4) A minimum radius curve for retrieval of 2.2m or less;

- 5) A working temperature range of at least 0°C to +50°C; and
  - 6) A construction (externally) of stainless steel.
- (ix) Once the probe is introduced into the inclinometer and prior to the commencement of readings, the probe shall be allowed the appropriate time for its temperature to stabilize.

**G3.4.17 Crackmeters And Tiltmeters**

- (a) Adequate crackmeters and tiltmeters shall be provided to monitor closely the transformation of cracks and angular displacement in surrounding buildings, key utilities and road structures.
- (b) Crack monitoring shall be carried out for existing cracks and new cracks.
- (c) The tiltmeter system includes a tilt plate, a portable tiltmeter, and a readout unit. Tilt plates can be either ceramic or bronze. Both shall be dimensionally stable and weather-resistant. Protective covers shall be provided for tilt plates. The portable tiltmeter shall use a force-balanced servo-accelerometer to measure inclination. The accelerometer shall be housed in a rugged frame with machined surfaces that facilitate accurate positioning on the tilt plate. The bottom surface is used with horizontally-mounted tilt plates and the side surfaces are used with vertically-mounted tilt plates.

**G3.4.18 Vibration Monitoring**

- (a) Vibration monitoring shall be carried out on all buildings/structures adjacent to piling, excavation, soil improvement, and pipe jacking or tunnelling works. Vibration sensors shall be able to measure in three directions.
- (b) Vibration monitoring shall be carried out where the construction activity involves processes which are likely to result in significant vibrations (for example: piling, excavation, pipe jacking and tunnelling works, etc.) and where the vibration could adversely affect adjacent buildings, structures, utilities, instruments or equipment.
- (c) Where buildings, structures, utilities, instruments or equipment that are particularly sensitive to vibration (for example: oil storage tanks, substations, laboratories, hospitals, utilities, and structures on shallow foundation), vibration monitoring shall be carried out continuously when any construction activity is carried out near to these sensitive buildings, structures, tanks, utilities, instruments or equipment.

**G3.4.19 Prisms**

- (a) For distance up to 50 m, mini-prism of approximately 25 mm visible diameter for short-range with precision glass prism ground to 2 arc seconds shall be installed. For distance greater than 50 m, large prism with approximately 60mm visible diameter shall be installed for long-range measurement. The prism shall be copper coated or un-coated with built-in filter installed to protect against condensation on the rear of the prism. The prism shall be mounted in air tight and robust housing for protection against moisture, shock and dirt.

### **G3.4.20    Extensometers**

- (a)    Magnetic Extensometers
  - (i)    Magnetic extensometers employed in the Works shall consist of a series of magnetic rings ("spider magnets") fixed into a maximum 150mm diameter borehole with a plastic tubing allowing access to the magnetic points for the measurement of settlement.
  - (ii)    The prongs on the magnetic rings shall be capable of extending beyond the circumference of the borehole into the surrounding soil.
  - (iii)    The complete downhole assembly shall be grouted in place ensuring that the access tube and magnets remain undisturbed during grouting operations. The access tube shall be capped to prevent ingress of ground water.
  - (iv)    The magnetic rings shall be housed in splay PVC holders suited to the ground into which they shall be installed to the acceptance of the S.O..
  - (v)    When the tubing is anchored in stable ground, a datum magnetic ring shall be set at its base. If the bottom of the tubing is not in stable ground, the depths of the magnets shall be referenced to a level survey point at the top of the tubing which shall be surveyed by precise levelling before readings are taken.
  - (vi)    The magnetic rings shall be installed at the depths accepted by the S.O. or at maximum 3m intervals.
  - (vii)    The reading system shall be reliable and require minimal maintenance over the required monitoring period. Thermal or other influences shall be negligible.
  - (viii)    The system shall have an accuracy of +1.0mm.
- (b)    Rod Extensometers
  - (i)    Rod extensometers shall be able to measure relative movements of both elongation and reduction in the length between the anchor and the reference collar to the acceptance of the S.O..
  - (ii)    A range adjustment device fitted at the reference collar shall extend the reading range beyond that of the measurement device.
  - (iii)    No more than six rods shall be installed in a single hole with each rod of a different length so that displacements at various borehole depths may be recorded.
  - (iv)    Where more than six rods are required another borehole shall be prepared as close as possible to the initial hole and rods set at alternate depths in each hole.
  - (v)    Each rod shall be individually isolated by its own plastic sleeve and the complete assembly shall be grouted in place fixing the anchors to the ground but allowing free movement of each rod within its sleeve.
  - (vi)    A single reference housing shall receive all of the rods from a drill hole and provide protection to the reference head.
  - (vii)    A level survey point shall be fixed to the top of the reference head of the extensometer to allow precise levelling back to a datum.

- (viii) Rods shall be fabricated from stainless steel or fibreglass.
- (ix) Anchor points shall have extendable prongs or be cadmium plated deformed mild steel bar of 22mm diameter and 500mm long.
- (c) Tape Extensometer
  - (i) A tape extensometer shall comprise a steel tape, portable measuring instrument with metric dial calliper and a pair of anchor clips
  - (ii) Anchors shall be located and fixed in a manner acceptable to the S.O..
  - (iii) The tape extensometer shall be capable of being read to +0.15 mm, and have a range of 20 m. A repeatability of reading to +0.15 mm shall be demonstrated prior to use.
  - (iv) Measurements shall be taken by connecting the tape hook to the pertinent eyebolt and the tape stretched across to the next bolt. The tension of the tape shall be adjusted and the tape and calliper readings taken.
  - (v) A minimum of two tape extensometers is to be provided at the commencement of work and these are to be calibrated against each other, so that in the event of damage to one tape, accuracy is not compromised.

#### **G3.4.21 Temperature Sensors**

- (a) The working range of temperature sensors shall be at least 0°C to +50°C with accuracy of  $\pm 1^\circ\text{C}$ . All sensors shall be capable of being remotely monitored.

#### **G3.4.22 Strain Gauges**

- (a) Strain gauges employed in the works shall be of vibrating wire type. Sensor must be securely attached by cable ties or other approved means.
- (b) Vibrating wire strain gauges shall have the following characteristics:
  - (i) Strain range  $2500 \times 10^{-6} \text{ m/m}$ ;
  - (ii) Accuracy  $\pm 3.0 \times 10^{-6} \text{ m/m}$ ; and
  - (iii) Resolution  $0.5 \times 10^{-6} \text{ m/m}$ .
- (c) Signal cable shall be shielded with four 22 gauge conductors with flexible polyurethane jacket or similarly approved. The cables shall be temperature rated to a minimum range of 0°C to +50°C.
- (d) Adequate protective measures shall be used to protect installed strain gauges from damage during the monitoring period.

#### **G3.4.23 Calibration**

- (a) All gauges and other instruments used for monitoring must be calibrated by a reputable testing company accredited by SAC-SINGLAS. Calibration certificates shall be kept at the Site and available for inspection by the S.O. at any time.
- (b) These gauges and other instruments used for calibration must be re-calibrated at appropriate intervals, and copies of the calibration certificates provided to the S.O..

**G3.4.24 Settlement/Deformations**

- (a) All monitoring instrumentation shall be designed to allow for any ground settlements, heaves or deformations which may occur.

**G3.4.25 Other Requirements**

- (a) All monitoring instrumentation shall be strong and durable, exhibit water tightness, corrosion resistance and long term stability to extreme temperature changes or gradient.

**G3.4.26 Testing and Commissioning**

- (a) Within one week of installation, the Contractor shall demonstrate to the S.O. that the instrumentation is functioning correctly. Such testing shall include, inter-alia:
  - (i) That the central tube of any piezometer or inclinometer is clear to the full design depth;
  - (ii) Rising and falling head permeability tests on piezometers; and
  - (iii) An initial survey of inclinometer tubes which will provide a baseline and also demonstrate that the tubes are clear.
- (b) The Contractor shall carry out a commissioning exercise to demonstrate to the satisfaction of the S.O. that all installed instruments are capable of providing reliable, sensible and repeatable data. Any instruments that fail the commissioning phase shall be abandoned and replaced by the Contractor.

**G3.4.27 Borelog Details**

- (a) Details of borehole logs, soil classification and laboratory test results from soil investigations along the same alignment are available for reference. The Contractor shall at his own expense conduct further soil investigations where necessary.

**G3.4.28 Data Management**

- (a) The Contractor shall submit full details of the proposed data management system for the acceptance of the S.O.. These shall include proposed formats for data acquisition, checking, processing and subsequent reporting to the S.O. for all instruments.
- (b) The Contractor shall establish a logical reference system for all the instrumentation equipment prior to installation and to the satisfaction of the S.O. so that records for any particular location can easily be recovered for interpretation or review.
- (c) When recording instrumentation readings, all site conditions that may affect the results shall be recorded. These shall include temperature and humidity readings, instrument type, location reference, datum and time of reading, personnel carrying out the readings and measuring, and instrument or readout unit reference shall also be recorded. Comments shall be recorded where observations of unusual conditions could influence the instrumentation or the structure.

- (d) Reporting of monitoring results shall be submitted to the S.O. on a weekly basis (with progress site photos attached) unless specified otherwise in an acceptable format including a plan showing the location of the structure and instrumented lines and shall comprise at least the following:
  - (i) Plots of measured parameters versus time, depth or location of tunnel, etc. as agreed with the S.O. (e.g. settlement versus time, horizontal deflections versus time, load versus depth of excavation, settlement versus location of tunnel/depth of excavation etc.);
  - (ii) Comparison of data with pre-determined trigger values; and
  - (iii) Recommendations on actions, if any, to be taken.
- (e) Interpretative reports shall be issued at weekly intervals based on the factual information received but when established trends indicate potential problems then such areas of concern shall be re-appraised at intervals consistent with the data collection and reports issued accordingly as required by the S.O.. These reports shall include a summary plot of all instrumented lines or points which indicate a trend (including historical data where appropriate) that could result in a trigger level being exceeded and any instrumented line or point in which a trigger level has been exceeded. The Contractor shall also submit his recommendations for any protective measures or actions that are necessary. A summary of the interpretative reports in a format acceptable to the S.O. shall be provided in the Contractor's Monthly Report.
- (f) On expiry of the Defects Liability Period, the Contractor shall be required to prepare and submit 3 sets of final reports in approved format to the S.O.. All reads out units and their accessories shall also become the property of the Contractor on completion of the Defects Liability Period. The Contractor shall also fill all boreholes, demolish all structures of any monitoring instruments above ground and reinstate the areas where the instruments were sited to their original conditions.
- (g) It shall be clearly understood that the submission of the report of monitoring to the S.O. are for the S.O.'s information. The Contractor shall be solely responsible for the analysis and interpretation of all the readings and measurements and for taking all rectification measures where necessary. Where abnormal readings or measurements are obtained, the Contractor shall immediately stop all activities that may worsen the soil/geotechnical movements or changes and shall immediately investigate the cause of abnormalities. The Contractor shall take all necessary rectification measures before resuming such activities, all at the Contractor's own cost and at no other expense to the Board.

#### **G3.4.29 Protection and Maintenance of Monitoring Instrument**

- (a) Every practical measure shall be taken to prevent damage to the instruments and ancillary equipment during handling, installation and subsequent operations. Any instrument and/or ancillary equipment damaged during handling, installation or subsequent operation shall be replaced immediately. Cabling and tubing shall be snaked or coiled to provide sufficient slack to accommodate potential ground movement.

- (b) The instruments shall be removed from site by the Contractor upon completion of the Works or as and when directed by the S.O.. All costs incurred for the removal of the instruments shall be deemed to have been included in the rates for the items on instrumentation in the Bills of Quantities. All read out units and the instruments shall remain the property of the Contractor.

**G3.4.30 Assessment of Damage to Buildings and Utilities**

- (a) Both temporary and permanent works shall be designed to ensure that ground movements are kept to an absolute minimum. In addition, proven construction techniques shall be used and good workmanship provided in restricting ground loss and water table drawdown.
- (b) The prediction of the ground movements due to bored tunnelling and excavations shall be based on the latest Land Transport Authority's Civil Design Criteria Specification for Road and Rail Transit Systems.
- (c) All buildings and structures within the assessment zones shall be assessed for damage. The methodology for assessment of potential damage to buildings or structures shall be based on the latest Land Transport Authority's Civil Design Criteria.
- (d) The Contractor shall notify the S.O. and the Authority of any breach of Alert and/or Action Levels. Failure to do so may result in the revoke of the Permit to Commence Works and actions will be taken for any non-compliance.

**END OF SECTION**